

URL Shortener — Module Dependencies

Project: URL Shortener

Technology: Python, FastAPI, SQLAlchemy, SQLite, Pydantic

Location: /home/dominic/Documents/fastapiurl

Report #: 16/20

```
main.py → database.py → SQLite
      → models.py
```

Single module. No internal sub-modules.

Module Dependencies

Import Graph

```
main.py
├─ database.py → SQLAlchemy
├─ auth.py → jose, passlib
├─ shortener/router.py → models, schemas
├─ blog/schema.py → models, strawberry
├─ chat/router.py → models, openai
├─ queue/router.py → models, asyncio
└─ rag/router.py → models, sklearn
```

Dependency Matrix

Module	Depends On	Used By
app.main	auth, database	All routers
auth	database	main, all routers
shortener	models, auth	main
blog	models, auth	main
chat	models, auth	main
queue	models, auth	main
rag	models, auth	main

Circular Dependencies

None found.

External Library Dependencies

```
# requirements.txt
fastapi>=0.115.0
sqlalchemy>=2.0.0
pydantic>=2.0.0
python-jose[cryptography]>=3.3.0
passlib[bcrypt]>=1.7.4
httpx>=0.27.0
python-multipart>=0.0.9
uvicorn[standard]>=0.30.0
```

Internal Module Coupling

Module	Incoming Edges	Outgoing Edges	Cohesion
app.main	0	6	High (hub)
auth	1	2	Medium
shortener	2	1	Low
blog	2	1	Low

Package Health

Package	Version	Latest	Status
fastapi	0.115.0	0.115.x	
sqlalchemy	2.0.x	2.0.x	
pydantic	2.0.x	2.x	
python-jose	3.3.x	3.3.x	
passlib	1.7.x	1.7.x	

Dependency Recommendations

1. Update to latest patch versions for security fixes
2. Consider replacing python-jose with PyJWT for active maintenance
3. Add dependency pinning in requirements.txt
4. Use pip-audit for vulnerability scanning